



**Centre for
Healthcare Engineering and Learning**



HEAL A-THON 2026

10-hour State Level Healthcare Hackathon

SUBMISSION TEMPLATE



TEAM

HEALTH TECH

- PARTICIPANT 1- VARSHA PATTANSHETTI, ISE 6TH SEMESTER
- PARTICIPANT 2- SUHAN, ISE 6TH SEMESTER
- PARTICIPANT 3- ABHIRAM ADARSH BHAT, ISE 6TH SEMESTER

PROBLEM STATEMENT

Hemoglobin and Glucose testing today is invasive and dependent on dedicated medical hardware, limiting accessibility and frequent monitoring.

Project Bio-Echo overcomes this by:

- Using a smartphone camera and flash to capture Transillumination PPG signals
- Applying signal processing and machine learning to estimate Hb and Glucose
- Eliminating needles and additional hardware
- Enabling low-cost, software-based health screening

Bio-Echo converts a hardware-dependent diagnostic process into a scalable smartphone solution.

TECH STACK

- **Python** – Core programming language used for signal processing, machine learning modeling, and overall system logic.
- **OpenCV** – Used for capturing video frames and extracting RGB (red channel) intensity from the finger-over-camera recording.
- **NumPy** – Used for numerical computations and handling time-series PPG signal data.
- **SciPy** – Applied for bandpass filtering to remove motion noise and unwanted frequencies.
- **Scikit-learn** – Implemented regression models to estimate Hemoglobin and Glucose levels.
- **LightGBM / Random Forest** – Used for optimized regression performance and improved prediction accuracy.

WORKFLOW

DEVELOPMENT PLAN DURING HACKATHON (NOT PROJECT WORKFLOW)

STEP 1	STEP 2	STEP 3	STEP 4
Signal Acquisition (Hour 1–2) 1. Finger-over-camera video using smartphone flash 2. Extract red-channel intensity frame-by-frame 3. Generate raw PPG signal	Signal Processing (Hour 3–4) 1. Apply bandpass filter to remove noise 2. Use FFT to detect heart rate frequency 3. Stabilize and normalize waveform	Model Development (Hour 5–7) 1. Extract waveform features (peaks, amplitude, frequency) 2. Train regression model 3. Estimate Hemoglobin & Glucose values	Interface & Testing (Hour 8–10) 1. Develop simple clinical report interface 2. Display waveform + Hb & Glucose values 3. Add basic interpretation message 4. Test and prepare demo



ORIGINALITY



How is our solution novel and creative?

- Transforms a smartphone camera and flash into a non-invasive optical sensing system
- Integrates Transillumination PPG with signal processing and machine learning
- Provides a software-based alternative to traditional hardware-dependent diagnostics

Present Market Competitors:

- Glucometers – Require invasive finger-prick blood samples
- Laboratory Hemoglobin tests – Depend on clinical infrastructure
- Wearable devices – Limited to basic vital parameter monitoring

What makes Bio-Echo different:

- Needle-free estimation approach
- No dedicated medical hardware required
- Cost-effective and easily scalable through smartphones

SCALABILITY

Given 24 Hours, We Would Add:

- Cloud-based patient data storage to securely maintain and access individual health records
- Trend analysis feature to monitor Hemoglobin and Glucose variations over time
- Automated alert system to notify users when readings fall outside normal clinical ranges

How Scalable Is the Project?

- Designed to function on existing smartphones without requiring additional hardware
- Can be deployed as a mobile application for wide accessibility
- Cost-effective solution suitable for large-scale community and public health screening programs

Why Would Investors Be Interested?

- Addresses the widespread and growing burden of anemia and diabetes
- Software-based approach significantly reduces infrastructure and operational costs
- Opportunity to collaborate with healthcare providers, hospitals, and public health initiatives